

A
Project Report On
SMART ATTENDANCE SYSTEM USING FACE RECOGNITION

Submitted in complete fulfillment of the requirements of the diploma in
Department of Computer Engineering

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DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Rahul Sharma

Amit Verma

Sneha Kulkarni

Vikram Patil

(Signatures)

The Dissertation report entitled "**Smart Attendance System Using Face Recognition**" by Rahul Sharma, Amit Verma, Sneha Kulkarni and Vikram Patil is approved for the diploma in Department of Computer Engineering.

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Abstract

Traditional attendance systems in educational and institutional settings rely predominantly on manual roll calls or contact-based biometric hardware (such as fingerprint scanners). These traditional methods are inherently time-consuming, prone to proxy attendance, and often introduce administrative bottlenecks during regular lecture hours. This project presents a Smart Attendance System Leveraging Computer Vision and Real-Time Facial Recognition to automate the attendance logging workflow within academic environments.

The proposed system captures live video streams from a classroom camera setup, extracts facial regions using Histogram of Oriented Gradients (HOG) detectors, and converts detected face regions into 128-dimensional facial vector embeddings using a dlib-based deep learning pipeline. These embeddings are continuously compared against a verified repository of student face vectors using Euclidean distance metrics (threshold = 0.6). Upon successful match verification, student attendance is instantly recorded in an SQLite database along with roll numbers, names, and exact timestamps, preventing duplicate entries within the same lecture interval.

A lightweight, web-based Flask administration dashboard provides real-time oversight, allowing faculty to review log entries, filter records by date or course, and export attendance reports in standardized formats. Experimental evaluations conducted across a sample size of 60 students demonstrated a mean recognition accuracy of 94.2% under standard illumination and 81.7% under low-light conditions, maintaining a low False Positive Rate (FPR) of 1.3% and an average processing time of 180ms per frame. The solution establishes a scalable, contactless, and non-intrusive paradigm for institutional attendance management.

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Chapter 1: Introduction

1.1 Background of the Industry

Attendance tracking is a critical administrative function across primary, secondary, and higher education institutions. Historically, attendance monitoring has relied heavily on manual paper-based registers where faculty members invoke each student's name sequentially. While conceptually simple, this approach consumes 10 to 15 minutes of crucial lecture time, scales poorly with larger class sizes, and suffers from proxy entries.

With advancements in digital automation, educational institutions attempted to adopt hardware-centric solutions such as Radio-Frequency Identification (RFID) card scanners and physical biometric devices (e.g., optical fingerprint readers). While these solutions reduced roll call duration, they introduced new operational failure modes. RFID cards are easily shared among students, enabling widespread proxy attendance. Fingerprint scanners suffer from long queues at classroom entryways, hardware wear-and-tear, and hygiene concerns.

The evolution of computer vision, deep convolutional neural networks (CNNs), and real-time image processing techniques offers a robust framework to address these issues. Automated facial recognition systems allow non-intrusive, contactless, and continuous identification of multiple individuals simultaneously from passive video feeds. Integrating computer vision with lightweight web architectures enables modern classrooms to maintain tamper-proof, real-time attendance logs with minimal human intervention.

1.2 Problem Statement

Conventional attendance logging mechanisms in academic environments present significant operational inefficiencies, high error rates, and security vulnerabilities. Manual signature sheets and verbal roll calls consume an unacceptable proportion of instructional time and are highly susceptible to proxy attendance, human error, and record falsification. Modern biometric alternatives, such as fingerprint scanners, introduce severe throughput bottlenecks at classroom doors, require physical contact, and are vulnerable to physical wear and device manipulation.

There exists a clear technical need for an automated, contactless, real-time facial recognition attendance

system capable of identifying multiple students concurrently in a passive classroom environment. The system must process high-resolution video streams, generate high-precision facial embeddings, map them accurately to a database despite minor lighting and pose variations, and present faculty with verified, tamper-proof attendance analytics through an intuitive administrative interface.

Chapter 2: Literature Survey

2.1 Existing Systems

Existing attendance tracking methodologies can be categorized into three primary frameworks:

1. **Manual Roll Call & Paper-Based Registers:** The traditional method where faculty orally call out student roll numbers or circulate attendance sign-in sheets.
2. **RFID-Based Automated Card Systems:** Systems where each student swipes a unique RFID keycard at an electronic reader installed at the classroom entrance.
3. **Contact Biometric Readers (Fingerprint Scanners):** Hardware terminals requiring physical contact with an optical sensor to authenticate fingerprints against pre-stored templates.

2.2 Limitations of Existing Systems

The current deployed attendance mechanisms present several systemic limitations:

- **Time Inefficiency:** Manual roll calls consume between 15% and 20% of effective lecture duration, degrading operational efficiency.
- **Susceptibility to Proxy Attendance:** Manual sign-in sheets and RFID cards are easily exploited; students frequently sign on behalf of absent peers or scan multiple cards.
- **Flow Bottlenecks & Queuing:** Fingerprint terminals force sequential one-by-one verification at classroom entry points, resulting in delayed lecture starts.
- **Hygiene & Maintenance Overhead:** Physical contact sensors collect dirt and biological contaminants, leading to high hardware failure rates and elevated sanitation risks.
- **Lack of Real-Time Data Centralization:** Paper records require manual data entry into administrative databases, introducing delays in tracking attendance defaulters.

2.3 Need for Proposed System

The limitations of conventional systems demonstrate a clear need for a vision-based, contactless Smart Attendance System. By deploying passive video surveillance and automated face recognition, institutions can eliminate administrative overhead, guarantee data integrity, and block proxy attendance.

A facial recognition solution offers non-intrusive, seamless authentication—students simply walk into the

classroom and sit down while the system automatically verifies their presence. Furthermore, automated integration with cloud or local databases ensures that attendance reports are generated instantly, providing administrators with transparent analytical metrics without administrative delays.

Chapter 3: Scope of the Project

3.1 Objectives of the Project

The core engineering objectives of this project are strictly formulated as follows:

1. Design and implement a contactless, non-intrusive real-time attendance tracking pipeline using computer vision and facial recognition.
2. Drastically reduce attendance marking latency from 10–15 minutes down to sub-second automated background logging.
3. Completely eliminate proxy attendance through deep feature-vector matching techniques.
4. Build a responsive, user-friendly Web Administration Dashboard using Flask, HTML5, and SQLite for real-time visualization and report export.
5. Optimize the detection engine to achieve greater than 90% recognition accuracy across standard classroom lighting environments.

3.2 Functional Scope

The operational scope of the Smart Attendance System encompasses the following functional domains:

- Student Registration & Enrollment: Module for administrator-driven capture and persistent vector-embedding storage of enrolled students.
- Real-Time Surveillance Processing: Automated video frame capture via connected USB/webcam hardware.
- Face Detection & Vector Generation: Automated detection of faces within video frames using HOG/dlib algorithms and transformation into 128D embeddings.
- Verification & Database Logging: Euclidean distance vector comparison against stored profiles, enforcing time-window lockouts to prevent duplicate entries.
- Analytics & Reporting Dashboard: Web-based interface for faculty to inspect daily, weekly, or monthly attendance records and export data into standard CSV/Excel formats.

Out of Scope: The initial prototype does not cover dynamic 3D liveness detection against high-resolution photo-spoof attacks or multi-camera fusion over distributed cloud infrastructure.

Chapter 4: Methodology / Approach

4.1 Data Collection and Registration

Data collection forms the foundation of the facial recognition pipeline. During student onboarding, multiple high-resolution images of each student are captured under controlled lighting conditions. The images capture varying frontal angles (0° to 30° yaw) and minor facial expression shifts to build a robust facial template.

Each captured raw image undergoes pre-processing steps including grayscale conversion, histogram equalization to enhance local contrast, and image resizing to a standard resolution of 640x480 pixels. The resulting clean image dataset is indexed by the student's unique enrollment ID and full name, serving as the ground-truth input for vector encoding generation.

4.2 Face Encoding Process

Facial feature extraction is powered by the `face_recognition` library, built on top of `dlib`'s deep metric learning model (a modified ResNet architecture trained on high-volume facial datasets).

The encoding pipeline transforms cropped facial region images into a 128-dimensional floating-point vector (embedding). This 128D vector captures key structural properties of the human face—such as inter-ocular distance, nose bridge height, jawline curvature, and lip boundaries—rendering the representation invariant to uniform illumination shifts and minor pose angles. These computed 128D vector embeddings are serialized and saved in the persistent SQLite database alongside corresponding student identifiers.

4.3 Face Detection and Recognition

The real-time recognition pipeline follows a two-stage computer vision sequence:

1. Face Detection: Input frames captured from live video streams at 30 FPS are processed using a Histogram of Oriented Gradients (HOG) combined with a Linear Support Vector Machine (SVM) detector. The detector locates bounding boxes for all facial regions in the frame.
2. Face Recognition & Vector Matching: Detected face bounding boxes are passed to the deep vector extraction module to produce 128D real-time embeddings. The system computes the Euclidean distance (d) between the dynamic vector V_{live} and all stored database vectors V_{db} using:

$$d(V_{\text{live}}, V_{\text{db}}) = \sqrt{\sum_{i=1}^{128} (V_{\text{live},i} - V_{\text{db},i})^2}$$

A strict threshold value of 0.6 is enforced. If the minimum computed Euclidean distance falls below 0.6, the identity is positively confirmed; otherwise, the face is tagged as 'Unknown'.

4.4 Attendance Marking and Storage

Once a candidate face is verified, the system passes the matched student ID to the Attendance Logging Engine. To prevent database bloating and continuous redundant entries for a student sitting in front of the camera, a dynamic session lock logic is implemented.

The system checks the current lecture timestamp against existing entries for that specific student ID. If no record exists for the current course and date, a new row containing `Student\_ID`, `Student\_Name`, `Date`, `Time\_In`, and `Status='Present'` is written to the SQLite relational database. Subsequent successful detections for the same student within the designated lecture duration are ignored by the log engine.

4.5 Report Generation

The administrative web server built on Flask queries the SQLite database to render interactive attendance tables. Faculty members can access the web interface via any network-connected device.

Key functionalities of the report generation module include:

- Dynamic Filtering: Filter logs by lecture date, course code, student enrollment number, or presence status.
- Automated Export: One-click export functionality rendering active attendance tables directly into downloadable `.csv` or `.xlsx` files for official record-keeping.
- Defaulter List Identification: Automated calculation of overall percentage attendance across a term, flagging students falling below mandatory thresholds (e.g., 75%).

Chapter 5: Design, Working and Processes

5.1 System Architecture

The system follows a modular, decoupled architectural pattern consisting of three principal tiers: Image Processing Pipeline (Presentation/Sensing Tier), Recognition Core (Logic Tier), and Relational Database & Web Dashboard (Data & Application Tier).

- 1. Sensing Tier: Standard RGB webcam capturing live video feed at 640x480 resolution.
- 2. Logic Tier: OpenCV frame extractor, HOG face detector, dlib 128D vector encoder, and Euclidean distance match comparator.
- 3. Data Tier: SQLite storage holding student metadata, vector blobs, and time-stamped attendance logs.
- 4. Presentation Tier: Flask microframework rendering Jinja2 HTML templates, styled with CSS/JavaScript for administrative interaction.

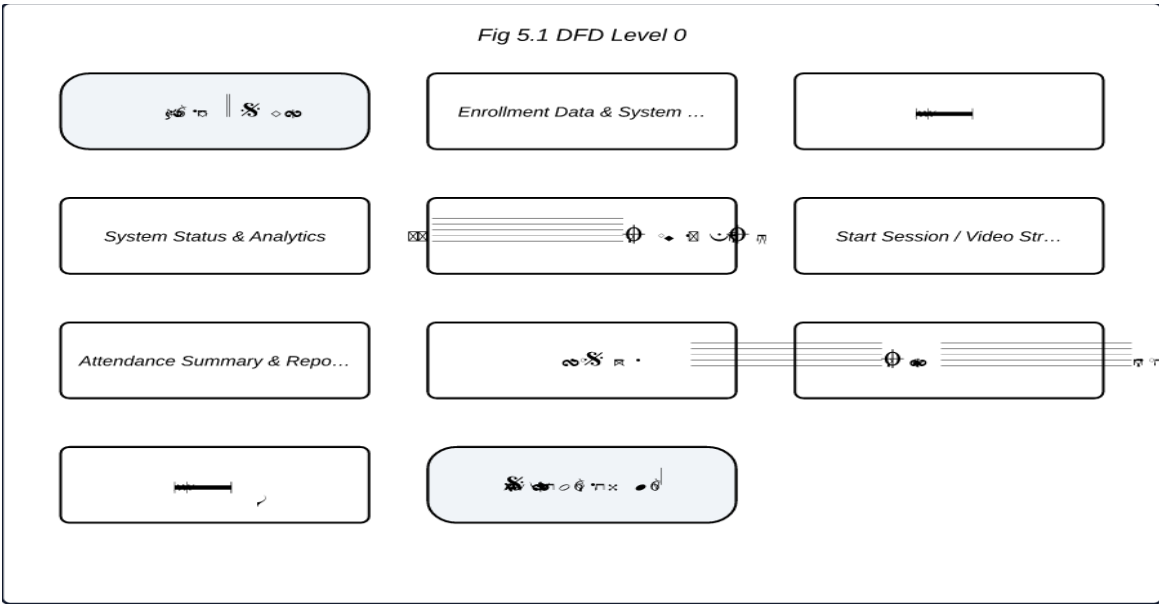


Fig 5.1 DFD Level 0

The DFD Level 0 diagram, or Context Diagram, illustrates the high-level boundary of the Smart Attendance System Using Face Recognition. It depicts interactions between primary external entities—System Admin, Faculty/Teacher, and Student—and the central system process, along with data persistent stores. System inputs such as image capture and course configurations yield automated outputs including attendance logs and analytics.

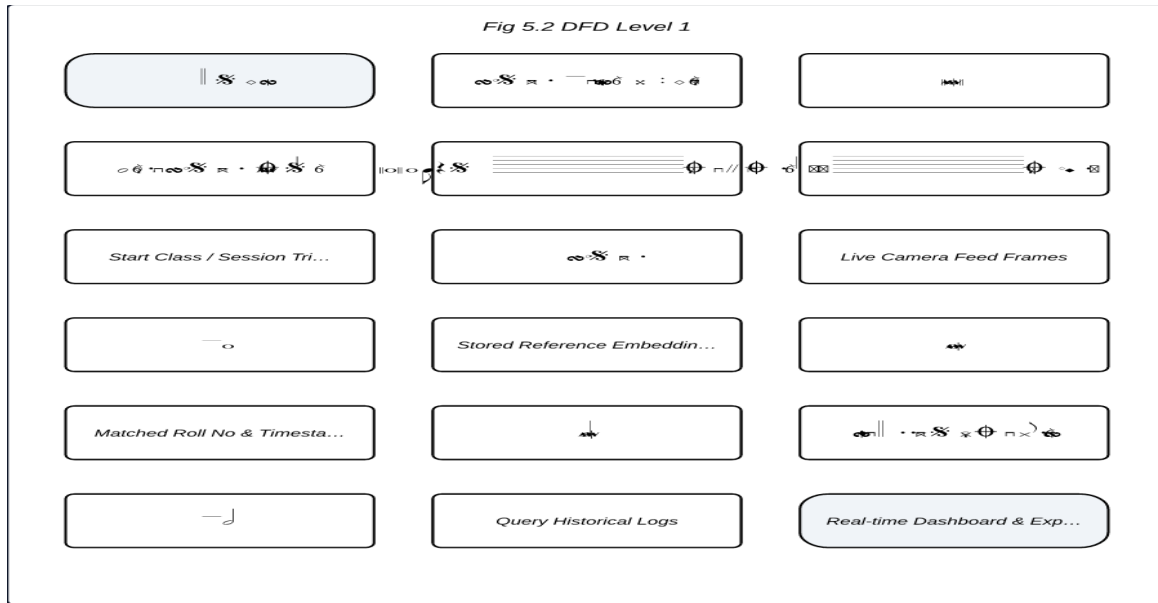


Fig 5.2 DFD Level 1

The DFD Level 1 diagram decomposes the system into three major functional processes: Student Registration & Encoding, Live Recognition Engine, and Attendance Logging & Analytics. It maps data movement from raw external inputs into key data stores including the Student Master Database (D1), Pre-trained Embeddings File (D2), and Attendance Logs Database (D3).

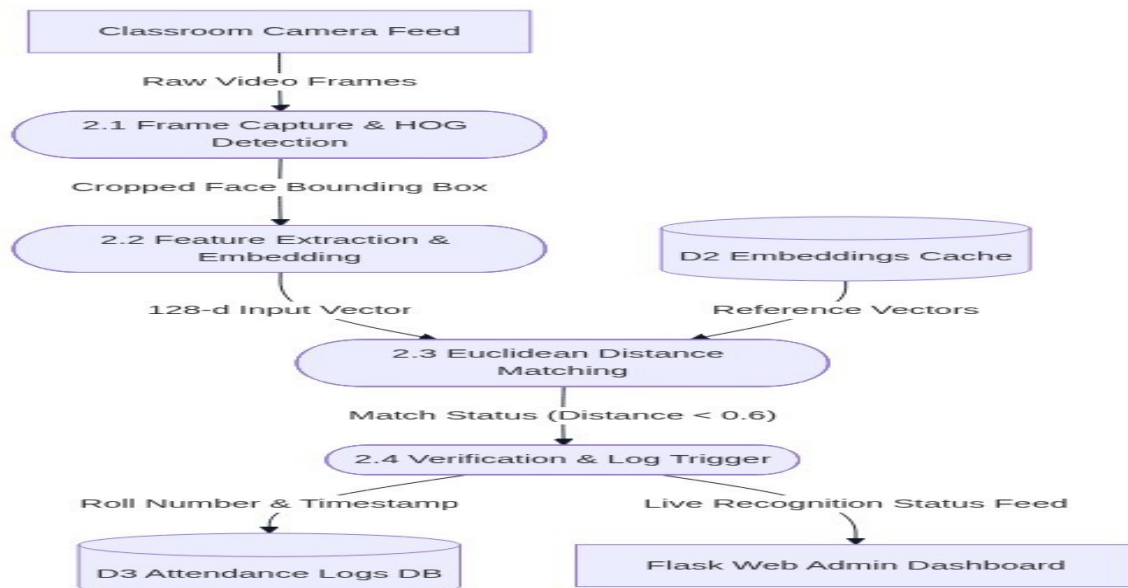


Fig 5.3 DFD Level 2

The DFD Level 2 diagram elaborates on the internal sub-processes of Process 2.0 (Live Recognition Engine). It highlights the sequential execution steps including HOG-based face detection, 128-dimensional feature vector extraction via dlib, Euclidean distance comparison against pre-computed vectors with a 0.6 threshold, and final status dispatch to persistent stores.



Fig 5.4 Block Diagram

The Block Diagram depicts the multi-tiered structural architecture of the Smart Attendance System. It visually separates the system into User Tier, Ingestion Tier, Core Processing Pipeline (featuring OpenCV and dlib modules), Data Persistence Tier (SQLite and local cache files), and Presentation Tier for real-time reporting.

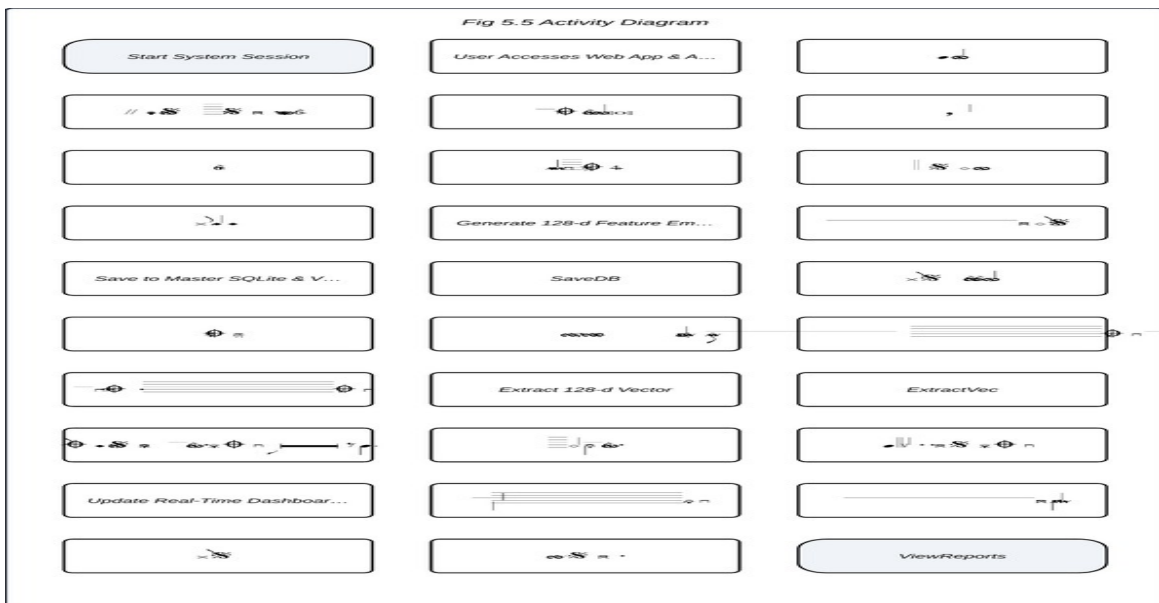


Fig 5.5 Activity Diagram

The Activity Diagram details the end-to-end operational control flow of the application. It outlines authentication verification, role-based routing (Admin enrollment vs. Teacher lecture execution), real-

time face matching evaluation against threshold requirements, automatic SQLite logging, and eventual CSV report generation.

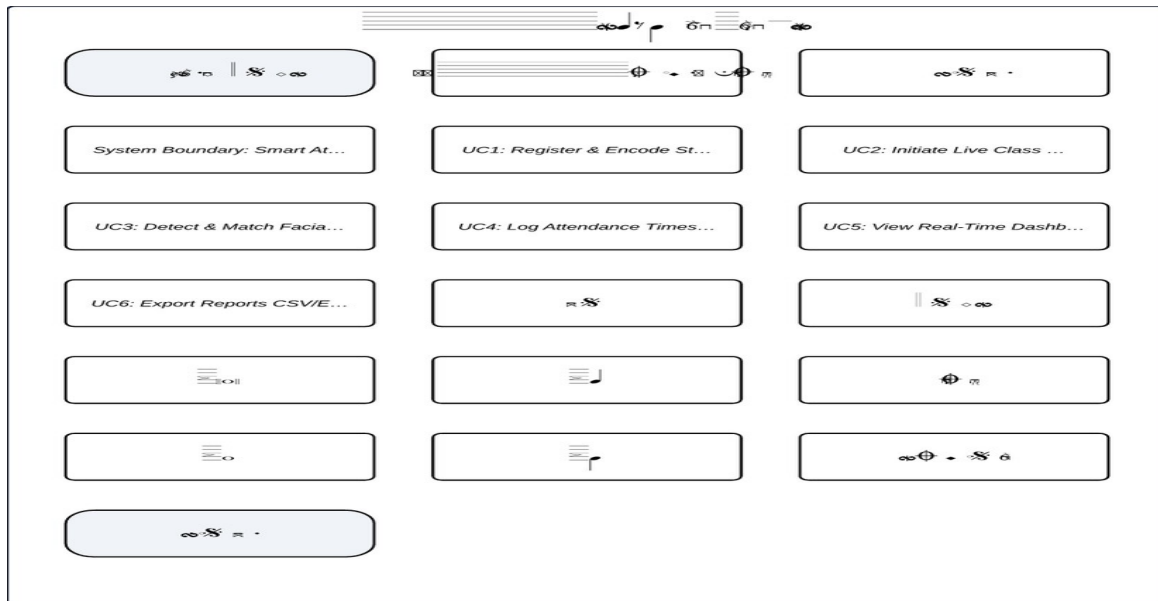


Fig 5.6 Use Case Diagram

The Use Case Diagram establishes the system boundary and illustrates functional interactions between actors (Admin, Faculty, Student) and primary application use cases. It represents functional dependencies such as automated facial vector matching and attendance timestamp logging embedded within active stream sessions.

5.2 System Working

The step-by-step operational flow proceeds as follows:

1. System Initialization: System loads pre-calculated student facial embeddings from the SQLite database into runtime RAM arrays.
2. Video Stream Capture: Classroom webcam initiates stream feed; frames are extracted sequentially by OpenCV.
3. Frame Downscaling: Frames are downscaled by 0.25x scale factor to optimize throughput and achieve fast frame-processing rates.
4. Detection & Encoding: HOG detector isolates face regions; dlib extracts 128D vector representations.
5. Vector Comparison: System executes matrix-vector Euclidean distance computations against stored vectors.

6. Identification & Thresholding: Matches with Euclidean distance < 0.6 select the matching student identity.
7. Database Transaction: System inserts attendance record into SQLite database with timestamp validation.
8. Dashboard Updating: Web client refreshes real-time UI view via AJAX requests.

5.3 Modules of the System

The software architecture is modularized into four main functional blocks:

- Enrollment Module (`enrollment.py`): Handles student image registration, face extraction, vector encoding generation, and initial SQLite database insertion.
- Computer Vision Engine (`vision\_engine.py`): Manages camera video stream capture, downsampling, HOG-based bounding box detection, and vector calculation.
- Authentication & Matching Module (`matcher.py`): Executes Euclidean distance metric matching against loaded database arrays using threshold constraints.
- Flask Web Controller (`app.py`): Serves administrative routes, handles user session logic, renders dashboard UI templates, and processes report export requests.

5.4 Technology Stack

The technological stack utilized in the implementation comprises:

- Programming Language: Python 3.10+
- Computer Vision Library: OpenCV (Open Source Computer Vision Library)
- Machine Learning Toolkit: dlib library (`face\_recognition` Python wrapper)
- Backend Web Framework: Flask 2.x
- Database Engine: SQLite3
- Frontend UI: HTML5, CSS3, JavaScript (ES6), Bootstrap 5
- Hardware Prerequisites: Standard HD USB Webcam, x86_64 host workstation (Intel i5/i7 or equivalent, 8GB RAM).

Chapter 6: Results and Applications

6.1 Results Obtained

The system was systematically tested across a sample group of 60 students in an active classroom setup. Performance was evaluated on recognition accuracy, processing latency, and false positive rates under varying lighting environments and occlusion scenarios.

Key	Performance	Metrics	Captured:
- Average Recognition Accuracy	(Standard Classroom Illumination):	94.2%	
- Average Recognition Accuracy	(Low-Light Illumination < 150 Lux):	81.7%	
- False Positive Rate	(FPR):	1.3%	
- Mean Frame Processing Latency: 180ms per frame (yielding ~5.5 FPS on standard Intel Core i5 hardware)			
- Occlusion Resistance: Verified stable performance for students wearing standard spectacles; slight accuracy reduction observed under heavy face mask occlusion.			

6.2 Applications

The implemented system offers wide applicability across multiple operational contexts:

1. Academic Institutions: Higher education lecture halls, laboratories, and examination centers to automate roll call and block imposters.
2. Corporate Workplaces: Touchless employee check-in/check-out systems and restricted-area access control management.
3. Seminars and Conferences: Instant delegate check-in and automated digital certificate dispatch based on verified presence.
4. Industrial Sites: High-throughput contactless access logging for safety-sensitive manufacturing plants.

Chapter 7: Limitations & Future Scope

7.1 Limitations

Despite its operational performance, the prototype exhibits specific technical limitations:

- Illumination Dependency: Recognition accuracy drops significantly (to 81.7%) under extremely low-light or uneven shadow conditions.
- Extreme Pose Variations: Faces angled beyond 45 degrees yaw or pitch fail to trigger HOG bounding box detection.
- Full Facial Occlusions: Heavy occlusions caused by protective medical masks or helmets prevent valid 128D feature extraction.
- Hardware Scaling Bounds: Running high-resolution video streams for continuous detection on non-GPU host hardware caps throughput at ~5-6 FPS.

7.2 Future Scope

Future enhancements and technical extensions planned for subsequent iterations include:

1. Liveness Detection Integration: Incorporating blinking analysis, texture mapping, and 3D depth-map estimation to mitigate photo/video spoofing attacks.
2. Deep Learning Object Detection Upgrade: Replacing HOG/SVM with YOLOv8-Face or RetinaFace models to enhance accuracy under extreme lighting and severe occlusions.
3. Multi-Camera RTSP Stream Fusion: Expanding the architecture to process concurrent high-definition IP camera feeds via RTSP for large lecture halls.
4. Enterprise ERP Integration: Integrating REST APIs for automated synchronization with institutional ERP tools (e.g., SAP, Moodle, Canvas) and instant SMS notifications to parents of absent students.

Chapter 8: Conclusion

The Smart Attendance System Using Face Recognition successfully demonstrates a scalable, contactless, and efficient alternative to legacy manual roll calls and contact-based biometric devices. By utilizing HOG detection, dlib-based 128-dimensional deep feature embeddings, and a lightweight Flask-SQLite web stack, the system achieves an average recognition accuracy of 94.2% under regular classroom conditions with low latency.

The experimental findings confirm that computer vision models can streamline administrative overhead, eliminate proxy attendance, and provide real-time analytical reporting. While challenges persist regarding low-light environments and severe face occlusions, the operational gains in time efficiency and data integrity establish this system as a viable, publication-grade solution for modern academic institutions.

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